

CHAPTER 1: Introduction

INTRODUCTION TO ARTIFICIAL INTELLIGENT (AI)



REV: 20.02.2024

Introduction

INTRODUCTION TO ARTIFICIAL INTELLIGENT



Outline

- What is intelligence? The definition of Artificial Intelligence (AI)
- The Turing Test Approach
- The Foundations of AI
- The History of AI
- The State of the Art for AI
- The Future of AI

What is intelligence?

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- The definition of intelligence is **controversial**.
- **Intelligence** has been defined in many different ways, including one's capacity for [1]:
 - Logic
 - Abstract thought
 - Understanding
 - Self-awareness
 - Communication
 - Learning
 - Emotional knowledge (understanding and awareness of emotions)
 - Memory
 - Planning
 - Creativity
 - Problem-solving

What is AI?

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- AI is the science and engineering of making intelligent machines.
- The purposes of AI can be organized into four categories [2].

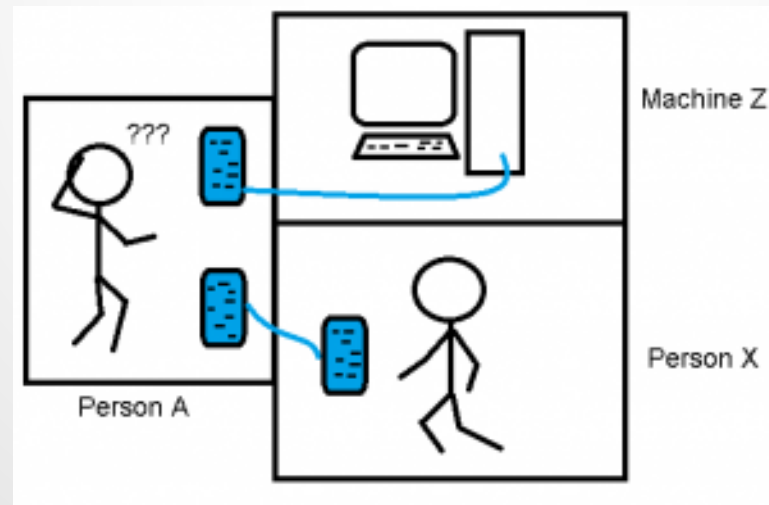
Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

Acting Humanly: The Turing Test

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- **The Turing Test**, proposed by Alan Turing in 1950, was designed to provide a satisfactory operational definition of intelligence. A computer passes the test if a human interrogator (Person A), after posing some written questions, cannot tell whether the written responses come from a person (Person X) or a computer (Machine Z).



The Turing Test [3]

The Turing Test Approach

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- A computer needs to possess the following capabilities to pass the Turing Test:
 1. **Natural language processing** to enable it to communicate successfully in English (or any other language).
 2. **knowledge representation** to store what it knows or hears.
 3. **automated reasoning** to use the stored information to answer questions and to draw new conclusions.
 4. **machine learning** to adapt to new circumstances and to detect and extrapolate patterns.
- If the perceptual abilities are tested, then the test is said to be the **Total Turing Test**. In order to pass the Total Turing Test, the computer also needs
 5. **computer vision** to perceive objects (to get input from the environment),
 6. **robotics** to manipulate objects and move about (to manipulate the environment).

The Turing Test Approach

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Has the Turing Test been passed?

- It is claimed that a computer program called Eugene Goostman passed the Turing test in 2014 at an event organized by the University of Reading. Here, the computer program simulated a **13-year-old** Ukrainian boy whose **second language** is English. Eugene managed to convince 33% of the human judges that it was human [4].

The Turing Test Approach

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Did Google pass the Turing Test? [5]

- In 2018, Google's CEO introduced their new voice technology, an artificial intelligence that called a hairdresser and successfully booked an appointment [5]. The sound of the AI voice was not different from a human voice. The woman who answered the phone seemed to have no idea she was talking to a computer [6]. The conversation was **confined to only an appointment**.
- This test did **not satisfy the conditions** of the Turing Test completely!

[5] <https://www.youtube.com/watch?v=zrJDZ7E7DbA>

[6] <https://bigthink.com/the-future/turing-test-imitation-game/>

The Turing Test Approach

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Did New Languages Models pass the Turing Test?

- Recently, several language models such as **ChatGPT**, developed by OpenAI, have become very popular. These represent an important milestone in the development of artificial intelligence. However, they did **not pass** the Turing Test!

The Turing Test Approach

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What Does the Turing Test show?

- The Turing Test shows that a machine can **act humanly** (also **act rationally**).
- But does it show that a machine **thinks like a human** or it **thinks rationally**?
- John Searle's **Chinese room argument** is the best-known thought experiment that opposed this idea (the machine that passes the Turing test thinks humanly or rationally).

The Chinese Room Experiment

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Chinese Room

The Chinese Room [7]

If you see this shape, "什麼" followed by this shape, "帶來" followed by this shape, "快樂"	then produce this shape, "爲天" followed by this shape, "下式".
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An instruction book [8]

- Some AI developers criticize the Chinese Room argument. They say the man does not know the Chinese but the system, including the room, man inside the room and instruction book, knows the Chinese.
- What is thinking? A human is really thinking?!! (freewill problem)

[7] <http://brianoverland.com/2015/05/20/a-i-and-consciousness/>

[8] http://www.mind.ilstu.edu/curriculum/searle_chinese_room/searle_chinese_room.php

The Foundations of AI

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The Foundations of AI

- **Computer Sciences** (How can we build more efficient computers? Programming, data structures, etc.)
- **Psychology** (How do humans and animals think and act?)
- **Philosophy** (Where does knowledge come from? Can formal rules be used to draw valid conclusions? How does knowledge arise from a physical brain?)
- **Mathematics** (Algorithms, probability, logic, etc.)

The Foundations of AI

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- **Economics** (Decision Theory, Game Theory)
- **Linguistics** (How does language relate to thought? Noam Chomsky's Theories...)
- **Neuroscience** (How do brains process information?)
- **Control Theory** (self-controlling machine)

The Short History of AI

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The Short History of AI

- **1943:** The first design of artificial neurons was made by McCulloch and Pitts. They proposed a model of artificial neurons in which each neuron is characterized as “on” or “off.”
- **1949:** Donald Hebb introduced the Hebbian learning rule for modifying the connection strengths between neurons.
- **1950:** Alan Turing’s article, titled “Computing Machinery and Intelligence”, was published. (He introduced the Turing Test, machine learning, genetic algorithms, etc.)

The Short History of AI

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- **1951:** Marvin Minsky and Dean Edmonds built the first neural network computer. They simulate a network of 40 neurons.
- **1958:** McCarthy introduced the high-level language **Lisp**, which was to become the dominant AI programming language for the next 30 years.
- **1959:** Herbert Gelernter (1959) developed the **Geometry Theorem Prover**, which was able to prove theorems that many students of mathematics would find quite tricky.
- **1970-1980:** The **Prolog** language, programming in logic, became popular in Europe.

The Short History of AI

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- In the 1970s, researchers were too optimistic about AI. For example, they claimed that **within 10-year**;
 - A computer would be a chess champion.
 - A significant mathematical theorem would be proved by a machine.
- U.S. National Research Council tried to speed up the translation of Russian scientific papers after the Sputnik launch in 1957. It was initially thought that simple syntactic transformations based on the grammar of Russian and English, and word replacement from an electronic dictionary, would be sufficient. However, this attempt **failed**, and in 1966 the translation project was canceled.

The State of the Art

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What can AI do today? A few examples:

- **Speech recognition:** A traveler calling United Airlines to book a flight can have the entire conversation guided by an automated speech recognition and dialog management system.
- **Game playing:** IBM's **DEEP BLUE** became the first computer program to defeat the world champion, Garry Kasparov, in a chess match. Kasparov said that he felt a “new kind of intelligence”. *Newsweek* magazine described the match as “**The brain's last stand.**”
- **Machine Translation:** Google Translator, ChatGPT, etc.

The State of the Art

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- **Spam fighting:** Each day, learning algorithms classify over a billion messages as spam.
- **Autonomous planning and scheduling:** A hundred million miles from Earth, NASA's Remote Agent program became the first onboard autonomous planning program to control the scheduling of operations for a spacecraft
- **Text classifiers:** Automatically classifying text according to their topic.
- **Robotic vehicles:** Google's driverless car is in the street now. The robots of Boston Dynamics make spectacular moves.

The State of the Art

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What can AI do today? Other examples...

- Optical character recognition
- Handwriting recognition
- Face recognition
- Diagnosis of diseases
- Artificial intelligence in computer games
- Navigation
- Antivirus programs
- Satellite control
- ...

The Future of AI

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- **Unemployment:** Will AI take place of workers? [9]

[9] <http://www.techinsider.io/researchers-predictions-future-artificial-intelligence-2015-10>

The Future of AI

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- Will AI make dream of human come true? [9]



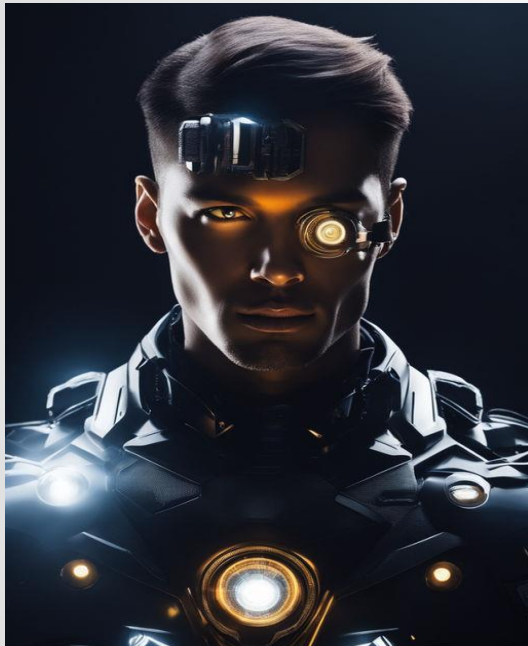
- Will AI destroy the world of people? [10]

[9] <http://www.techinsider.io/researchers-predictions-future-artificial-intelligence-2015-10>

[10] http://terminator.wikia.com/wiki/Terminator_Wiki

The Future of AI

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- **cybernetic organism:** human machine hybrid species [11]



- **The ethics of artificial intelligence:** The discussion about violation of humans? Or robot rights? [12]

[11] <https://playgroundai.com/search?q=Cyborg+android>

[12] <https://www.trustedreviews.com/news/meps-voting-on-whether-robots-should-have-rights-2949777>

Main Reference

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Main Reference

Stuart Russell, Peter Norvig; “Artificial Intelligence A Modern Approach”, Pearson, 3rd Edition, 2013.